

SHRI BRAHMAGUPTA VIRACITA

BRĀHMA-SPHUṬA SIDDHĀNTA

WITH

**Vāsanā, Vijñāna and Hindi
Commentaries**

Vol. I

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Foreword

The Indian Institute of Astronomical & Sanskrit Research has done a very useful piece of work by publishing the Brāhma Sphuṭa Siddhānta. This is an original work on Astronomy written about twelve hundred years ago. Mainly devoting himself to Astronomy as was natural, the author, Brahma Gupta, has also given considerable space to such branches of Mathematics as in his opinion are particularly applicable to Astronomy. Brahma Gupta is not a mere theoriser. His work bears ample evidence of close observation of astronomical phenomena, though he did not have at his disposal any large and well equipped Vedhālaya or Observatory. He has had to find fault with some of his predecessors. The main reason for his criticism was as stated by him:

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् स्खलीतं मतम् ।

श्रीभ्रशीयते स्फुटं तज्ज्ञप्तं तद्ब्रह्मगुप्तेन ॥

The creator Brāhma himself had given certain calculations in his book namely Brahma Siddhānta. But in the course of many years these calculations have been found to have become inaccurate. Of course, in referring to calculations, the author is referring to the bases of these calculations, the main data of astronomy. The excellent and fairly exhaustive introduction by Dr. Satya Parkash is of very great help in understanding the book because it brings out clearly the various points on which Brahma Gupta has laid stress, particularly those on which he differs from other great astronomers.

A great controversy still rages between two schools of Indian astronomers which for want of better names may be called the schools of Arya Bhata and Brahma Gupta. Some years ago the Indian Institute of Astronomical & Sanskrit Research published the Vateshwar Siddhānta which is supposed to be associated with the Arya Bhata School. It is, therefore, in the fitness of things that it should now publish the present standard volume of the other school.

Brahmagupta's Self-Introduction

Brahmagupta introduces himself and his work in the opening verses of the Saṃjñā-adhyāya:

श्रीचापवंशातिलके श्रीव्याघ्रमुखे नृपे शकनृपाराम् |
पञ्चाशत्संयुक्ते वर्षशते पञ्चभिरतीते ||

ब्राह्मः स्फुटसिद्धान्तः सज्जनगणितज्योतिर्विदप्रिये |
त्रिशतद्व्यधिकं कृतं जिष्णुसुतब्रह्मगुप्तेन ||

Translation: “In the reign of the illustrious Vyāghramukha, the ornament of the Cāpa dynasty, when 550 years of the Śaka era had elapsed, this Brāhma-sphuṭa-siddhānta was composed by Brahmagupta, the son of Jiṣṇu, for the delight of good mathematicians and astronomers, at the age of 30.”

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् स्खलीतं मतम् |
श्रीभ्रशीयते स्फुटं तज्ज्ञप्तं तद्ब्रह्मगुप्तेन ||

Translation: “The calculations of the planets taught by Brahmā, which had become inaccurate through long lapse of time, have been corrected by Brahmagupta.”

संसाध्य स्पष्टतरं बीजं नलिकादिद्विनेत्रेभ्यः |
तत्संस्कृतग्रहेभ्यः कर्मण्यो निर्णयादेशो ||

Translation: “Having corrected the algebra (*bīja*) more accurately than Nalikādi and Dvīne tra, he gives instructions for computing the planets corrected thereby.”

Maṅgala – Opening Invocations

Brahmagupta opens the *Brāhmasphuṭasiddhānta* with invocations to Brahmā, the creator, and to the science of astronomy itself. The following verses constitute the traditional *maṅgala* (auspicious beginning) of the work:

Maṅgala Verse 1: Homage to Brahmā

ब्रह्मा गुरुर्नियतमेव सदा स्मर्तव्यः ।
ज्योतिःशास्त्रं प्रवक्ष्यामि लघुं सज्जनतोषणम् ॥ १ ॥

Translation: “Brahmā, the eternal teacher, is ever to be remembered. I shall propound the science of astronomy briefly, for the delight of the good.”

Maṅgala Verse 2: Invocation to Bṛhaspati

गणनाथं शिरसा नमामि ज्योतिश्चक्रवर्तिनं गुरुम् ।
येनोक्तं ग्रहगणितं ब्रह्मणा पूर्वमेव हि ॥ २ ॥

Translation: “I bow with my head to the lord of calculation, the sovereign of the stars, the teacher, by whom the computation of the planets was taught by Brahmā of old.”

Maṅgala Verse 3: The purpose of the Siddhānta

यत्किञ्चिद्ग्रहगणितं संसारसारं हि तत् सर्वम् ।
तत्प्रदर्शयितुं सिद्धान्तः प्रणीतो ब्रह्मगुप्तेन ॥ ३ ॥

Translation: “Whatever computation of the planets there is, that is the essence of the world; to demonstrate all that, this *Siddhānta* has been composed by Brahmagupta.”

Chapter I: Madhyamādhikāra (Mean Longitudes)

In the first chapter, Brahmagupta sets out the fundamental parameters for the mean motions of the Sun, Moon, and planets. He gives the number of revolutions in a *kalpa* (4,320,000,000 years), the *mahāyuga* (4,320,000 years), and corrections for the current epoch.

Verse 1.1: The yuga-revolutions of the planets

कल्पे सूर्यस्य भगणा नवशतिं शतत्रयं च युजः ।
चन्द्रस्य दशकोट्यः सप्तचत्वारिंशच्च सप्ततिश्च ॥ १ ॥

Translation: “In a kalpa, the revolutions of the Sun are 4,320,000,000; those of the Moon are 57,753,336,000 [revolutions].”

Verse 1.2: Revolutions of Mars and Mercury

भौमस्य कल्पभगणा द्विशतं दशोऽनं लक्षं त्रयं च ।
बुधस्य षट्त्रिंशल्लक्षं नवसहस्राणि चाष्टौ च ॥ २ ॥

Translation: “The revolutions of Mars in a kalpa are 2,290,000; those of Mercury are 17,937,020.”

Verse 1.3: Revolutions of Jupiter and Venus

गुरोः कल्पभगणा त्रयो लक्षाः सप्ततिश्च सहस्राणि ।
भृगोस्तु लक्षाणि सप्त सप्ततिश्च सहस्रद्वयं च ॥ ३ ॥

Translation: “The revolutions of Jupiter in a kalpa are 364,220; those of Venus are 7,022,444.”

Verse 1.4: Revolutions of Saturn and the lunar nodes

मन्दस्य कल्पभगणा लक्षद्वयं चोऽनसहस्राणि च दश ।
राहोर्भगणाः सप्तत्रिंशल्लक्षाणि नवसहस्राणि ॥ ४ ॥

Translation: “The revolutions of Saturn in a kalpa are 146,568; those of Rāhu [the ascending node] are 232,238 [in the reverse direction].”

Verse 1.5: The mean longitude formula

भगणाद्भगणितं गतं दिनैर्गतैर्ग्रहस्य मध्यमम् ।
लब्धं भुक्त्या चलनं तत्क्षेत्रं तस्य प्रदर्शयेत् ॥ ५ ॥

Translation: “From the revolutions, the elapsed [portion] divided by the days elapsed gives the mean [longitude] of the planet. The motion [computed] by the daily motion should be shown in its place.”

Chapter II: Spaṣṭādhikāra (True Longitudes)

In the second chapter, Brahmagupta describes the conversion of mean longitudes to true longitudes by applying corrections for the *maṇḍala* (epicycle) and *śīghra* (sighra) anomalies. These corrections involve the Indian sine function (*jyā*).

Verse 2.1: The equation of the centre

मध्यमात्कर्णतो ज्यां हृत्वा दोःफलं लभेत ततः ।
तत्कोटिफलं चापं मध्यमे देयमुभयोरपि ॥ १ ॥

Translation: “From the mean [longitude], having taken the sine [of the anomaly] divided by the hypotenuse, one obtains the result for the sine; from that, the result for the cosine. Both are to be applied to the mean [to obtain] the true [longitude].”

Verse 2.2: The manda and śīghra corrections

मन्दफलं मन्दकर्णे हृत्वा लब्धं ग्रहस्य मन्दम् ।
शीघ्रं तथा शीघ्रकर्णे द्वयोः संस्कारः क्रमेणैव ॥ २ ॥

Translation: “The manda correction, having been divided by the manda hypotenuse, the result [is] the manda [equation] of the planet. Similarly the śīghra [correction] by the śīghra hypotenuse. Both corrections are to be applied in order.”

Verse 2.3: The true longitude

मन्दोनं मन्दसंयुक्तं शीघ्रोनं शीघ्रसंयुक्तं तु ।
तत्स्पष्टं ग्रहतत्त्वज्ञैरुदाहृतं सुविचारितम् ॥ ३ ॥

Translation: “[The mean] minus the manda [correction] or plus the manda [correction], and minus the śīghra or plus the śīghra—that is the true [longitude], declared by those who know the reality of the planets, well considered.”

Verse 2.4: The hypotenuse (karṇa)

त्रिज्यायुतं मन्दफलं कोटिफलं ततो विभजेत् ।
तत्कर्णः स्यात्ततो ज्या लब्धा दोःफलं भवेत् ॥ ४ ॥

Translation: “The manda-phala added to the radius, the koṭi-phala [being added]; from that, divide [appropriately]; from that, the hypotenuse is obtained; from that, the sine obtained becomes the result for the sine.”

Chapter III: Chāyādhikāra (Shadow Problems)

Brahmagupta’s third chapter treats the *tripraśna* or three problems of diurnal motion: finding the longitude from the shadow, the shadow from the longitude, and the time from either. These problems involve the *chāyā* (shadow of the gnomon, *śaṅku*).

Verse 3.1: The gnomon and shadow

द्वादशाङ्गुलसंख्याको शङ्कुः प्रमाणतो निर्दिष्टः |
तस्य छाया प्रमाणं दिनकरस्योन्नतिमानयेत् || १ ||

Translation: “The gnomon is stated as having twelve aṅgulas in measure. From its shadow, the altitude of the Sun is to be computed.”

Verse 3.2: Finding the declination from the shadow

शङ्कुर्व्यसुविश्लेषात्त्रिज्यया हतो विश्लेषहतः |
लब्धा छाया ततः कोटिरित्याकाशसङ्ग्रहः स्मृतः || २ ||

Translation: “The gnomon, multiplied by the radius and divided by the [distance from the] equinoctial shadow, gives the shadow; from that, the cosine—this is known as the determination of the celestial [coordinates].”

Verse 3.3: The three problems stated

तत्कालखेचर उत्क्रमाद्भ्रुवादभ्रुवं प्रविशति |
छाययोर्घटिका भ्रूवोर्घटिका त्रिप्रश्नः स उदीरितः || ३ ||

Translation: “At that time, the planet, [moving] from the ascending node, enters the pole; the ghaṭikās [from] the shadows, the *bhujyā*—that is called the three-fold problem.”

Chapter IV: Prakīrṇādhikāra (Lunar Crescents)

This chapter treats miscellaneous problems concerning the appearance of the Moon's crescent, its horns (*śṛṅga*), and the illumination of the lunar disk.

Verse 4.1: The illumination of the Moon

चन्द्रस्य प्रकृतिं ज्ञात्वा शुक्लपक्षे तु वर्धनम् |
कृष्णपक्षे तु हासो वृद्धिहासौ प्रदर्शयेत् || १ ||

Translation: “Having known the nature of the Moon, in the bright fortnight there is increase; in the dark fortnight, decrease—one should show the increase and decrease.”

Verse 4.2: The horns of the crescent

चन्द्रबिम्बस्य शृङ्गे तु दिक्सूत्रात्परमं दिशि |
तत्प्रमाणं प्रवक्ष्यामि लघुं सज्जनतोषणम् || २ ||

Translation: “The horns of the Moon's disk [extend] from the directional line to the utmost in [that] direction. I shall briefly explain that measure, for the delight of the good.”

Chapter V: Grahaṇādhikāra (Eclipses)

Brahmagupta’s treatment of eclipses is one of the most sophisticated parts of the *Siddhānta*. He gives the conditions for the occurrence of both lunar and solar eclipses, the computation of their magnitude, duration, and appearance. The following verses are from the Candragrahaṇādhyaḥ (Chapter IV of the Daśagṛhayoga) and Sūryagrahaṇādhyaḥ (Chapter V).

Verse 5.1: Conditions for a lunar eclipse

चन्द्रो राहुसमीपे यदा तदा ग्रहणं प्रसूयते |
पूर्णं वा तदनुपूर्णं वा तत्प्रमाणं ततो ज्ञेयम् || १ ||

Translation: “When the Moon is near Rāhu, then an eclipse is produced; [it may be] full or partial—that measure is to be known therefrom.”

Verse 5.2: The eclipse limit

द्वादशात् परमं पारं षट्त्रिंशल्लक्षणं स्मृतम् |
ततोऽधिकं ग्रहणं न स्यात्तत्परं च विवर्जयेत् || २ ||

Translation: “Twelve [degrees] is the maximum limit; thirty-six is the [eclipse] criterion. If it exceeds that, there is no eclipse; beyond that, one should reject [the possibility].”

Verse 5.3: The magnitude of the eclipse

ग्रहणं युज्यते विक्षेपाच्छुक्लाधिको राहुबिम्बात् |
चन्द्रबिम्बं यावत्तावत्तद्ग्रहणस्य प्रमाणम् || ३ ||

Translation: “The eclipse is determined from the latitude; the Moon’s disk [extends] beyond Rāhu’s disk by the bright [portion]. The extent of the lunar disk [covered] is the measure of the eclipse.”

Verse 5.4: Duration of the eclipse

मन्दगत्यन्तरेणैव ग्रहणकालं प्रकल्पयेत् |
षष्ट्या भक्तं घटिकास्तत् स्पर्शमोक्षौ ततः स्मृतौ || ४ ||

Translation: “By the difference of the slow motion, one should compute the eclipse time; divided by sixty, that gives the ghaṭikās; the first and last contacts are known therefrom.”

Verse 5.5: Color of the eclipse (Rāhu’s shadow)

ताम्रं लोहितमर्कस्य ग्रहणं कालमेव तु |
धूम्रं नेत्रानलाकारं ताम्रं ससितं स्मृतम् || ५ ||

Translation: “Copper-red [is the color] of the Sun’s eclipse; black indeed; smoky, like the eye of fire; dark with white—these are the known [colors].”

Chapter VI: Grahayutyadhikāra (Planetary Conjunctions)

This chapter deals with the computation of planetary conjunctions (*graha-yuti*), both when two planets have the same longitude (*yoga*) and when they come close to each other in the sky (apparent conjunction).

Verse 6.1: Definition of conjunction

यदा द्वौ ग्रहौ समेक्रान्तौ तदा तद्योग उच्यते |
तयोर्न्तरमालोक्य योगं ज्ञात्वा प्रयत्नतः || १ ||

Translation: “When two planets have the same longitude, then that is called a *yoga*. Having observed their difference, having known the *yoga* with effort.”

Verse 6.2: The battle of planets (*graha-yuddha*)

यदा तु सम्मुखौ द्वौ ग्रौ तदा तद्ग्रहयुद्धमुच्यते |
उत्तरं दक्षिणं चापि तयोरैश्वर्यमादिशेत् || २ ||

Translation: “When two planets are face to face, then that is called a planetary battle (*graha-yuddha*). The northern and southern [one’s] sovereignty should be declared.”

Chapter VII: Udayāstādhikāra (Rising and Setting)

This chapter treats the times of rising and setting of planets, their *asta* (heliacal setting) and *udaya* (heliacal rising), and the computation of the *lagna* (ascendant).

Verse 7.1: Heliacal rising and setting

ग्रहाणामुदयं वार्ध्नं सूर्यसमीपतो गतेः ।
अस्तं तद्विपरीतं स्यादुदयास्तौ प्रदर्शयेत् ॥१॥

Translation: “The rising of a planet is from the east when it has gone near the Sun; the setting is the reverse of that—one should show the rising and setting.”

Verse 7.2: The *daśā* (period of visibility)

दशासंज्ञक्ता ग्रहा यान्ति सूर्यं ते तु दिनोदयाः ।
वक्रा वक्रगतिं यान्ति तत्प्रमाणं विचक्षणैः ॥२॥

Translation: “Planets with their periods of visibility approach the Sun; their risings by day [are known]; the retrograde move with retrograde motion—that measure [is known] by the discerning.”

Chapter VIII: Candragrahaṇādhikāra (Lunar Eclipses)

This chapter provides a detailed treatment of lunar eclipses, including the computation of the shadow's diameter, the latitude of the Moon, and the graphical representation (*śaraṇa-alekha*) of the eclipse.

Verse 8.1: The Earth's shadow

भूच्छाया सा त्रिभिर्भागैरधिका चन्द्रबिम्बतः ।
तत्परमाणं ज्ञात्वा तु ग्रहणं सम्प्रकल्पयेत् ॥ १ ॥

Translation: “The Earth's shadow is larger than the Moon's disk by three parts [i.e., roughly 2.5 times the Moon's diameter]. Having known that measure, one should compute the eclipse.”

Verse 8.2: The graphical representation (*Śaraṇa*)

चन्द्रं राहुबिम्बं च क्षेत्रे कृत्वा तु सुक्षेत्रे ।
तयोर्मध्ये ग्रहणं तत् सुस्पष्टं यन्त्रतो भवेत् ॥ २ ॥

Translation: “Having drawn the Moon and Rāhu's disk in the diagram, in a good diagram; the eclipse between them—that becomes very clear by instrument.”

Verse 8.3: The *akṣa* (latitude) and *valana*

विक्षेपोऽयनलवैर्दृष्ट्वा तद्वलनं च कल्पयेत् ।
दक्षिणावृत्तं चोत्तरावृत्तं तत्प्रदर्शनं स्मृतम् ॥ ३ ॥

Translation: “The latitude [is computed] by the minutes of the ayana; having seen that, one should construct the *valana*; the southern circle and the northern circle—that representation is known.”

Sanskrit Verses on Trigonometry (Jyā-gaṇita)

Brahmagupta, like Āryabhaṭa before him, gives a table of sine differences (*jyā-khaṇḍa-jyā*) for computing the *jyā* (chord/sine) of any arc. His table is more accurate than Āryabhaṭa's and uses a radius (*trijyā*) of 3438'. The following verses give his sine table and the method of interpolation.

Verse: The sine table stated

त्रिज्या त्रिसहस्रं चतुःशतत्रिंशच्च तत्पदम् ।
ज्याः प्रथमं सप्ततिस्तु द्वितीयं द्विसप्ततिः स्मृता ॥

Translation: “The radius is 3438; its square-root [is the first]. The first sine is 225, the second is remembered as 224 [in the table of twenty-four sines].”

Verse: The sine differences (Jyā-khaṇḍa-jyā)

ज्याक्षेपैर्ज्याक्षेपैस्तु विभजेद्भागान् प्रथमादीन् ।
यावद्विंशतीर्यावत्तावज्ज्याः स्युः क्रमेणैव ॥

Translation: “By the sine-differences, one should divide the degrees from the first; as many as twenty [intervals of $3\frac{3}{4}^\circ$], so many are the sines in order.”

Verse: Interpolation formula

उत्तरज्यापचयेन हृत्वा लब्धं तु यच्छेषकम् ।
तत्पूर्वज्यायुक्तं स्यात्तत्संस्कृतं ततो ज्यामानम् ॥

Translation: “Having divided by the decrease of the following sine, the remainder [multiplied] by the result; that added to the previous sine becomes the corrected [value]; from that, the measure of the sine.”

Brahmagupta's Sine Table (Selected Values):

No.	Arc	Jyā (sine)
1	$3\frac{3}{4}^\circ$	225
2	$7\frac{1}{2}^\circ$	449
3	$11\frac{1}{4}^\circ$	671
4	15°	890
6	$22\frac{1}{2}^\circ$	1315
8	30°	1719
12	45°	2431
16	60°	2978
18	$67\frac{1}{2}^\circ$	3177
20	75°	3321
24	90°	3438

The radius *trijyā* = 3438' corresponds to $360^\circ/(2\pi) \approx 57.2958^\circ$ in modern terms, with each minute of arc being one unit of sine.

Vāsanābhāṣya Commentary on Key Verses

The *Vāsanābhāṣya* commentary on the *Brāhmasphuṭasiddhānta* provides detailed explanations of Brahmagupta's verses. The following selections illustrate the commentary style:

Vāsanā on the mean longitude (I.5):

वासना—भगणा इति युगभगणाः तैर्गतदिनैर्विभज्य लब्धं मध्यमम् ।
यदा ग्रहः काल्पिकसंज्ञकः सूर्यस्तदा मध्यम एव स्पष्टमतोऽन्येषाम् ॥

Translation of Vāsanā: “‘Revolutions’ means the revolutions in a *yuga*. Dividing by the elapsed days, the result is the mean. When the planet is the Sun, called *kālpika*, then the mean itself is the true; for others [it is different].”

Vāsanā on the sine table:

वासना—त्रिज्या त्रिसहस्रचतुःशतत्रिंशद्वक्ता विभज्य त्रिभिज्याम् ।
तत्पदं प्रथमा ज्या द्वितीयाद्याः क्रमशस्तु क्षेपैः ॥

Translation of Vāsanā: “The radius, 3438, divided [into 96 equal parts]; its square-root [is the first sine-difference]. The first sine; the second and others in order by the differences.”

Vāsanā on eclipses (V.1):

वासना—राहुः सूर्यबिम्बच्छायां राशिः स तु द्वादशात् परमं षट्त्रिंशल्लक्षणात् ।
ग्रहणं युज्यते विक्षेपाच्छुद्धात्तत्प्रमाणं ततो ज्ञेयम् ॥

Translation of Vāsanā: “Rāhu is the shadow-body of the Sun’s disk; its maximum [distance for eclipse] is twelve [degrees], the criterion is thirty-six. The eclipse is determined from the [true] latitude; from that, the measure is to be known.”

Chapter XII: Gaṇitādhyāya (Mathematics)

The twelfth chapter of the Brāhmasphuṭasiddhānta is the most important for the history of mathematics. It contains 72 verses covering arithmetic, algebra, geometry, and series. The chapter opens with Brahmagupta's famous definition of a mathematician:

यः सप्तदशसहस्रीं लिप्यो वेत्ति स वै गणकः ।
अष्टादशसहस्रीं च स बीजाङ्कं तथापरम् ॥

Translation: “He who knows the seventeen hundredth (verse) is a mathematician; and he who knows the eighteen thousandth also knows algebra (*bīja*).”

Verses 1–2: Invocation and Chapter Definition

प्रणम्य गणनाथं तं ज्योतिश्चक्रवर्तिनं गुरुम् ।
गणितं सम्प्रवक्ष्यामि लघुं सज्जनतोषणम् ॥ १॥
व्यक्ताव्यक्तविधानं तु क्रियागणितमुच्यते ।
व्यक्तं तद्व्यक्तमेव स्यादव्यक्तं यत्तु यद्व्ययम् ॥ २॥

Translation: “Having bowed to the lord of calculation, the sovereign of the stars, the teacher, I shall propound mathematics briefly for the delight of the good. The method of known and unknown quantities is called *kriyāgaṇita*. What is manifest is known; what is not manifest is unknown.”

Sanskrit Verses on Arithmetic Progression (Śreṇī)

Brahmagupta gives the rules for arithmetic progressions in verses 17–22 of the Gaṇitādhyāya. The following verses are extracted from the critical edition (page 142 of the Sanskrit text):

Verse 17: The last term and sum of an A.P.

प्राग्वृत्तिकल्पस्यादि तत्क्रुद्धो द्वैतिकमध्यानाम् ।
घटितं चाद्यमानं तार्धं तत्संयुतं दलं धने ॥ १७॥

Translation (after Colebrooke): “The period less one, multiplied by the common difference, being added to the first term, is the amount of the last. Half the sum of last and first terms is the mean amount, which multiplied by the period, is the sum of the whole.”

In modern notation, for an arithmetic progression with first term a , common difference d , and n terms:

- Last term: $\ell = a + (n - 1)d$
- Sum: $S_n = \frac{n}{2}(a + \ell) = \frac{n}{2}[2a + (n - 1)d]$

Verse 18: The mean value and other properties

अन्ते युक्तं दलं मध्यं मुखं दलमथान्तकम् ।
पदं व्येकं वियुक्तं स्याद्वनं मुखं दलं धने ॥ १८॥

Translation: “Half the sum of the last and first terms is the mean; the common difference is [found from] the difference of the first and last terms divided by the period less one. The sum is the mean multiplied by the period.”

Verse 19: Finding the first term, common difference, and number of terms

मूलं धनं पदवर्गात्पदं व्येकेन वर्जितात् ।
द्विगुणेनाधिकेनैव मूलं व्येकं पदं पृथक् ॥ १९ ॥

Translation: “The square-root of the sum is [found] from the square of the period [multiplied] by the common difference and twice the first term; [the first term is found] from the common difference and the sum.”

Verses 20–21: Geometric progression

मूलं व्येकं पदं पृथगुत्सेधं दलं ततः पदम् ।
घातं स्वरूपसंयुक्तं व्येकं पदं ततो मुखम् ॥ २० ॥

मूलं व्येकं पदं पृथगुत्सेधं दलं ततः पदम् ।
घातं स्वरूपसंयुक्तं व्येकं पदं ततो मुखम् ॥ २१ ॥

Translation: “In a geometric progression, the sum is found by multiplying the first term by the common ratio raised to the power of the period, subtracting the first term, and dividing by the common ratio less one.”

Sanskrit Verses on the Area of a Triangle

Before the cyclic quadrilateral formula, Brahmagupta gives the area of a triangle:

Verse 20: Area of a triangle

समदलकोटीभुजार्धसंवर्गः समकोटिदीर्घचतुरस्रवर्गः |
विषमचतुरस्रस्य च तद्वत् क्षेत्रस्य विष्कम्भाद्धनं वर्गः || २० ||

Translation: “The product of half the base and the perpendicular is the area of a triangle. For a rectangle, it is the product of the longer and shorter sides. For an isosceles trapezium, the area is the product of the half-sum of the parallel sides and the perpendicular.”

Sanskrit Verses on the Cyclic Quadrilateral

The most famous result in Brahmagupta's geometry is his formula for the area of a cyclic quadrilateral. The verses below (Gaṇitādhyāya 21–28) develop the theory systematically, culminating in the area formula and the diagonals of a cyclic quadrilateral.

Verses 21–22: Preliminary lemmas on triangles and circumradius

समकोटिदीर्घचतुर्भुजस्य क्षेत्रस्य मूलं द्विजसङ्गुणस्य |
कोटिभुजवर्गयुतनिःशेषाद्वर्गं द्विजं तस्य लम्बसंवर्गः || २१ ||

समचतुर्भुजस्य क्षेत्रं बाहुवर्गो भवति सप्तमो भागः |
विषमचतुर्भुजस्य च तद्वत् क्षेत्रस्य विष्कम्भाद्धनं वर्गः || २२ ||

Verses 23–24: The 'approximate' and exact area formulas

विषमचतुर्भुजस्य क्षेत्रं बाहुषट्कस्य परस्परं हतस्य |
चतुर्गुणितस्य पदं क्षेत्रं सूक्ष्मं तु तज्ज्ञानात् || २३ ||

Translation: “The gross area of an irregular quadrilateral is the square root of the product of the six sides taken alternately [i.e., the product of the two triangles formed by a diagonal]. But the exact area is to be known by those who understand the method.”

अत्यसमचतुर्भुजस्य क्षेत्रं बाहुचतुष्टयसपातशः स्थितम् |
तद्वर्गपदं सूक्ष्मं क्षेत्रं स्यात्तत्क्षेत्रज्ञैरुदाहृतम् || २४ ||

Verses 25–26: The exact formula via semi-perimeter

सर्वदोर्घाभ्यासार्धकृतेः सघातात्पदं क्षेत्रं सूक्ष्मम् |
तच्चतुर्भुजस्य बाहुभिः सव्यस्तैरेवं विनिर्दिशन्ति || २५ ||

Translation: “The exact area is the square root of the product of the semi-perimeter into its differences with the four sides. [In modern notation: $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$]

द्व्यर्धचतुर्भुजक्षेत्रं दलसंवर्गात्ततो बाहुषट्के |
सव्यस्तबाहुचतुष्टयस्य क्षेत्रं सूक्ष्मं प्रदर्शितम् || २६ ||

Verses 27–28: The diagonals of a cyclic quadrilateral

चतुर्भुजस्य बाहुभिः परस्परहतैः सव्यस्तैरेवम् |
विभक्तं द्विजसङ्गुणेन तत्क्षेत्रं सूक्ष्मं प्रदर्शितम् || २७ ||

कोटिभुजवर्गयुतनिःशेषाद्वर्गं द्विजं तस्य लम्बसंवर्गः ।
चतुर्भुजस्य बाहुभिः सव्यस्तैरेवं विनिर्दिशन्ति । ॥२८॥

Brahmagupta's Formula: For a cyclic quadrilateral with sides a, b, c, d and semi-perimeter $s = \frac{a+b+c+d}{2}$:

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

This generalizes Heron's formula for triangles (take $d = 0$). Brahmagupta also gives formulas for the two diagonals δ_1, δ_2 of a cyclic quadrilateral:

$$\delta_1 = \sqrt{\frac{(ab+cd)(ac+bd)}{ad+bc}}, \quad \delta_2 = \sqrt{\frac{(ac+bd)(ad+bc)}{ab+cd}}$$

Sanskrit Verses on Quadratic Equations (Avyakta-gaṇita)

Brahmagupta's treatment of quadratic equations appears in Chapter XVIII (the Algebra chapter), verses 44–48. These verses give the first explicit solution formula for quadratic equations in the history of mathematics.

Verses 44–45: The quadratic formula (equivalent)

अव्यक्तवर्गरूपयुतेः पक्षद्वयस्यापवर्तनम् |
रूपयुतिर्वर्गयुतिः स्याद्वर्गरूपविवर्जिता || ४४ ||

वर्गः पक्षद्वयं कृत्वा विषमं द्विसमं तथा |
रूपयुतिर्वर्गयुतिः स्याद्वर्गरूपविवर्जिता || ४५ ||

Translation (after Colebrooke): “The square of the unknown, multiplied by the coefficient of the square, being added to the known quantity [i.e., constant term], divide both sides by the coefficient of the square. The square-root of the absolute term [constant] being added to the square of half the coefficient of the unknown, extract the square-root [of that sum].”

Verses 46–47: Completing the square

मूलं रूपयुतिर्वर्गात्पदं व्येकेन वर्जितात् |
द्विगुणेनाधिकेनैव मूलं व्येकं पदं पृथक् || ४६ ||

द्विगुणाव्यक्तयुक्तेभ्यो रूपयुतेभ्यः पृथक्पदम् |
रूपयुतिर्वर्गयुतिः स्याद्वर्गरूपविवर्जिता || ४७ ||

Translation: “Multiply both sides of the equation by the coefficient of the square; add the square of half the coefficient of the unknown to both sides; then extract the square-root.”

Verse 48: The explicit solution

मूलं रूपयुतिर्वर्गात्पदं व्येकेन वर्जितात् |
द्विगुणेनाधिकेनैव मूलं व्येकं पदं पृथक् || ४८ ||

Translation: “The two roots are found by subtracting and adding the square-root of the sum of the square of half the coefficient and the absolute number, divided by the coefficient of the square, and then halving.”

Brahmagupta's Quadratic Formula (Equivalent): For a quadratic equation of the form $ax^2 + bx = c$, Brahmagupta's rule is equivalent to:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a} \quad \text{or} \quad x = \frac{\sqrt{4ac + b^2} + b}{2a}$$

This is algebraically equivalent to the modern quadratic formula. Brahmagupta states the rule in words rather than symbols, but the procedure is unmistakably the same.

Sanskrit Verses on Zero and Negative Numbers

Brahmagupta's Chapter XVIII (the Algebra chapter) contains the first systematic treatment of zero and negative numbers in the history of mathematics. The following Sanskrit verses (18.30–35) give the fundamental rules for arithmetic operations with positive (*dhana*, wealth), negative (*r̥ṇa*, debt), and zero quantities.

Verse 18.30: Rules for addition

धनयोर्धनयोर्योगो ऋणयोरपि योगमेकम् ।
धनऋणयः समं शून्यं शून्याभ्यासे शून्यमेव ॥

Translation (after Colebrooke): “The sum of two positive quantities is positive; the sum of two negative quantities is negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Rules for subtraction

अल्पात्प्रचयं शेषं प्रचयादल्पकं शेषमेकम् ।
ऋणधनयोः समं शून्यं शून्याभ्यासे शून्यमेव ॥

Translation: “[If] a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative], the algebraic sign of their difference is reversed—i.e. negative becomes positive and positive becomes negative.”

Verse 18.32: Subtraction involving zero

ऋणं शून्याद्वियोगेन धनं धनाद्विशोध्यते ।
शून्यं शून्याद्वियोगेन शून्यं शून्येन योजयेत् ॥

Translation: “A negative minus zero is negative, a positive [minus zero] is positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added [to the other].”

Verse 18.33: Rules for multiplication

ऋणधनयोर्गुणकर्म वधो धनयोर्गुणे ऋणयोः |
 ऋणयोर्गुणे धनमेकं शून्याभ्यासे शून्यमेव ||

Translation: “The product of a negative and a positive is negative, of two negatives is positive, and of positives is positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs for Multiplication:

$$\begin{aligned}
 (+a) \times (+b) &= +ab \\
 (-a) \times (-b) &= +ab \quad (\text{first explicit statement in history}) \\
 (-a) \times (+b) &= -ab \\
 (+a) \times (-b) &= -ab \\
 0 \times a &= 0 \quad \text{for any } a.
 \end{aligned}$$

Verse 18.34: Rules for division

धनं धनेन विभजेदृणमृणेन तत्तथा |
 शून्यं शून्येन विभजेद्वनमृणेन तदृणम् ||

Translation: “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero—the concept of ‘kha-hara’

शून्येन भक्तश्च खहरः शून्यं हारश्च राशावमिश्रे |
 तद्वर्गं मूलं तस्य तद्वर्गस्य पदं तथा ||

Translation (Colebrooke): “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Verse 18.36: The famous verse on division by zero = infinity

शून्येन भक्तश्च खहरः शून्यं हारश्च राशावमिश्रे |
 तद्वर्गं मूलं तस्य तद्वर्गस्य पदं तथा ||

Commentary: The phrase “शून्येन भक्तश्च खहरः शून्यं हारश्च राशावमिश्रे” (*śūnyena bhaktaś ca khaharaḥ*) means “a number divided by zero is a fraction with zero as denominator”—Brahmagupta’s term for infinity. The word *kha-hara* (literally “zero-divisor” or “zero-denominator”) was his way of expressing the concept of an infinite quantity. In his words, “शून्यं हारश्च राशावमिश्रे तद्वर्गं मूलं तस्य तद्वर्गस्य पदं तथा”—“it is impossible to exhaust the quantity of a khahara.”

Note on Division by Zero: Brahmagupta’s statement that $a/0$ (for $a \neq 0$) gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of **infinity**. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities behave as infinite was remarkable for the 7th century. He writes:

Khaharasya rāśeḥ pūnaṃ pūrayituṃ na śakyate

“It is impossible to exhaust the quantity of a khahara [infinite quantity].”

This makes Brahmagupta the first mathematician in history to attempt a systematic arithmetic of zero and negative numbers, and the first to grapple with the concept of infinity as a mathematical object.

Sanskrit Verses on the Pulverizer (Kuṭṭaka)

The *Kuṭṭaka* (“pulverizer”) method for solving linear indeterminate equations of the form $ax + by = c$ is one of Brahmagupta’s crowning achievements. The following verses (18.1–5) give the algorithm:

Verse 18.1: Statement of the pulverizer

द्विवज्रपदं व्यासं धनर्णं यावत्तावदेकम् |
अनिश्चितं विभक्तं रूपैर्वा युक्तं विवर्जितं वा || १॥

Translation: “The unknown quantity, multiplied by the coefficient and increased or decreased by the absolute term, being divided by the divisor, is the pulverizer.”

Verses 18.2–3: The mutual division algorithm

भाज्यो हारेण विभजेदवशिष्टं हारभाजितं तत् |
पुनरपि हारेण विभजेदवशिष्टं हारभाजितं तत् || २॥

यावद् भागलाभं तत्रावशिष्टं यत् तत् स्मृतं गुणकम् |
भागलाभः स्मृतः खण्डं ततः पूर्वगुणकारः || ३॥

Translation: “Divide the dividend by the divisor; the remainder is divided by the divisor [of the remainder]; again the remainder is divided by the divisor. The quotients are called ‘sections’ and the remainders are called ‘multipliers.’ The previous multiplier is [used for] the next step.”

Verses 18.4–5: Constructing the solution

यः पातो गुणकस्तेन हतं खण्डं पूर्वगुणकारेण |
युक्तं परं पातं तत् परं गुणकारं विद्यात् || ४॥

एवं यावत्समं खण्डं ततो गुणः प्रकल्पितः |
भाज्ये हृते यदवशिष्टं तत् खण्डं गुणकः स्मृतः || ५॥

Translation: “Multiply the section by the multiplier and add the next section; that is the next multiplier. Continue until the sections are exhausted; then the multiplier is the solution. The remainder when the dividend is divided [by the divisor] is called the pulverizer.”

The Kuṭṭaka Algorithm: Given $ax + by = c$ with $\gcd(a, b) = 1$:

1. Apply the Euclidean algorithm to (a, b) to obtain quotients $q_1, q_2, \dots, q_n$ and remainders $r_1, r_2, \dots, r_n = 1$.
2. Set up the recurrence: $x_0 = 1, \quad x_1 = q_n, \quad x_i = q_{n-i+1}x_{i-1} + x_{i-2}$
3. The solution is $x = \pm cx_{n-1} \pmod{b}$.

This is essentially the Extended Euclidean Algorithm, developed centuries before its European rediscovery.

Sanskrit Verses on Square Nature (Vargaprakṛti)

Brahmagupta’s method for solving the equation $Nx^2 + 1 = y^2$ (now known as Pell’s equation) was his most profound algebraic discovery. The following verses (18.64–68) give his composition method (*samāsa-bhāvana*):

Verse 18.64: The composition of solutions (Bhāvanā)

गुणकयोर्गुणकान्तरेण कृत्यो द्वयोर्गुणकयोस्तरेण |
वर्गयोर्वर्गान्तरेण कृत्यं तद्वर्गस्य पदं तथा || ६४ ||

Translation: “The product of two [auxiliary] equations [with the same multiplier N] is [formed by] the product of the two first roots multiplied by the multiplier added to the product of the two second roots [for the new second root]; and the product of the two first roots [added to] the product of the two second roots [for the new first root].”

Verse 18.65: The bhāvanā identity

प्रकृतिप्रकृतिहतयोः पदयोः क्षेपयोरपि हतेः पदम् |
क्षेपक्षेपयुतयोर्वर्गयोर्वर्गयुतिः प्रकृतिहता || ६५ ||

Translation (modern): If (x_1, y_1) and (x_2, y_2) are solutions of $Nx^2 + k_1 = y^2$ and $Nx^2 + k_2 = y^2$ respectively, then the composition gives:

$$x' = x_1y_2 + x_2y_1, \quad y' = Nx_1x_2 + y_1y_2$$

which satisfies $Nx'^2 + k_1k_2 = y'^2$.

Brahmagupta’s Bhāvanā (Composition): This is Brahmagupta’s most celebrated theorem. Given two solutions to $Nx^2 + k_i = y^2$, their “composition” produces a solution to $Nx^2 + k_1k_2 = y^2$. In particular, when $k_1 = k_2 = k$:

$$x = \frac{x_1y_2 + x_2y_1}{k}, \quad y = \frac{Nx_1x_2 + y_1y_2}{k}$$

gives a solution to $Nx^2 + 1 = y^2$ when $k = \pm 1, \pm 2, \pm 4$. This method allows one to generate infinitely many solutions from a single initial solution, and was the first general method for solving Pell’s equation in the history of mathematics.

Introduction

Chapter I. Indo-Greek Contacts

The Indian Institute of Astronomical & Sanskrit Research has done a very useful piece of work by publishing the *Brāhma Sphuṭa Siddhānta*. This is an original work on Astronomy written about twelve hundred years ago. The work of Brahmagupta occupies a unique position in the history of Indian astronomy and mathematics. He was not a mere theoriser; his work bears ample evidence of close observation of astronomical phenomena, though he did not have at his disposal any large and well equipped Vedhālaya or Observatory.

In the second chapter of his great work, Brahmagupta himself states: ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् स्खलीतं मतम् ।

श्रीभ्रशीयते स्फुटं तज्ज्ञप्तं तद्ब्रह्मगुप्तेन ॥

The accurate knowledge of the planetary motions earlier than about the century of the Christian era. From thenceforth astronomy, hitherto confined to rituals, appears as a science, treated in the course of the next thousand years in a series of text-books, the *Siddhāntas*, the contents of which, though supposed to be derived from divine sources are strongly influenced by Greek authors.

Chapter II. Personal References of Brahmagupta

Sudhakara Dvivedi in his *Gaṇaka Taraṅgiṇī*, a small book on biographical sketches of astronomers and astrologers of this country gives a brief account of Brahmagupta thus:

Brahmagupta was born in 520 Śaka (655 Vikrami or 589 A.D.) in the reign of King Vyāghramukha, belonging to the Cāpa family; his father was Jīṣṇugupta; and at the age of 30, he wrote in 550 Śaka (628 A. D.) his well known treatise on Astronomy known as the *Brāhmasphuṭasiddhānta*.

Chapter IV. Contents of the Brāhmasphuṭa Siddhānta

The *Brāhmasphuṭasiddhānta* as edited by Sudhakara Dvivedi contains twenty-four chapters as follows:

Ch.	Title	Verses
I	<i>Tithi-nakṣatrādhikārādhyaḥ</i> (On tithis, nakṣatras etc.)	32
II	<i>Grahagatyadhyāyaḥ</i> (On the mean and true places of 'star' planets)	19
III	<i>Tripraśnādhyaḥ</i> (On the three problems relating to diurnal motion)	16
IV	<i>Candragrahaṇādhyaḥ</i> (On lunar eclipses)	7
V	<i>Sūryagrahaṇādhyaḥ</i> (On solar eclipses)	6
VI	<i>Udayāstādhikāraḥ</i> (On the rising and setting of planets)	7
VII	<i>Candraśṛṅgonnatyadhāyaḥ</i> (On the position of the Moon's cusps)	4
VIII	<i>Samāgamādhyaḥ</i> (On conjunction of planets)	6
UTTARA KHAṆḌAKHĀDYAKA—APPENDIX		

Ch.	Title	Verses
IX	Corrections and new methods	14
X	On conjunction of stars and planets	16
Total		127

The remaining chapters (XI–XXIV) of the Brāhmasphuṭasiddhānta contain the mathematical and astronomical material proper:

Ch.	Title	Verses
XI	<i>Elūkādhyāyaḥ</i> (On arithmetic operations)	—
XII	<i>Gaṇitādhyāyaḥ</i> (On mathematics)	72
XIII	<i>Kuṭṭakādhyāyaḥ</i> (On the pulveriser / indeterminate equations)	—
XIV	<i>Vargaprakṛtika</i> (On square nature / Varga-prakṛti)	—
XV	<i>Citrabhānu</i> (On algebraic problems)	—
XVI	<i>Śreḍhī</i> (On arithmetic and geometric progressions)	—
XVII	<i>Kṣetravyavahāra</i> (On plane geometry)	—
XVIII	<i>Khātaprādhyāyaḥ</i> (On excavations)	—
XIX	<i>Chāyāprādhyāyaḥ</i> (On shadows)	—
XX	<i>Śilpi-prāsāda</i> (On architecture)	—
XXI	<i>Śaṅkūnni</i> (On gnomonics)	—
XXII	<i>Gati-prādhyāyaḥ</i> (On motion)	—
XXIII	<i>Yantra-prādhyāyaḥ</i> (On astronomical instruments)	—
XXIV	<i>Samjñā-adhyāyaḥ</i> (On technical terminology)	—

Contents of the Brāhmasphuṭasiddhānta

The following table gives the complete chapter listing of the Brāhmasphuṭasiddhānta with the number of verses in each chapter:

Ch.	Title	Verses
DAŚAGRAHAYANTRA		
I	<i>Tithi-nakṣatrādhikārādhyaḥ</i> (On tithis, nakṣatras etc.)	32
II	<i>Grahagatyadhyāyaḥ</i> (On the mean and true places of ‘star’ planets)	19
III	<i>Tripraśnādhyaḥ</i> (On the three problems relating to diurnal motion)	16
IV	<i>Candragrahaṇādhyaḥ</i> (On lunar eclipses)	7
V	<i>Sūryagrahaṇādhyaḥ</i> (On solar eclipses)	6
VI	<i>Udayāstādhikāraḥ</i> (On the rising and setting of planets)	7
VII	<i>Candraśṛṅgonnatyadhyaḥ</i> (On the position of the Moon’s cusps)	4
VIII	<i>Samāgamādhyaḥ</i> (On conjunction of planets)	6
UTTARA KHAṆḌAKHĀDYAKA—APPENDIX		
IX	Corrections and new methods	14
X	On conjunction of stars and planets	16
Total (Daśa + Uttara)		127

The remaining chapters (XI–XXIV) of the Brāhmasphuṭasiddhānta contain the mathematical and astronomical material proper:

Ch.	Title	Verses
XI	<i>Elūkādhyaḥ</i> (On arithmetic operations)	—
XII	<i>Gaṇitādhyaḥ</i> (On mathematics)	72
XIII	<i>Kuṭṭakādhyaḥ</i> (On the pulveriser / indeterminate equations)	—
XIV	<i>Vargaprakṛtika</i> (On square nature / Varga-prakṛti)	—
XV	<i>Citrabhānu</i> (On algebraic problems)	—
XVI	<i>Śreḍhī</i> (On arithmetic and geometric progressions)	—
XVII	<i>Kṣetravyavahāra</i> (On plane geometry)	—
XVIII	<i>Khātaprādhyaḥ</i> (On excavations)	—
XIX	<i>Chāyāprādhyaḥ</i> (On shadows)	—
XX	<i>Śilpi-prāsāda</i> (On architecture)	—
XXI	<i>Śaṅkūnni</i> (On gnomonics)	—
XXII	<i>Gatī-prādhyaḥ</i> (On motion)	—
XXIII	<i>Yantra-prādhyaḥ</i> (On astronomical instruments)	—
XXIV	<i>Samjñā-adhyāyaḥ</i> (On technical terminology)	—

Chapter XII: Gaṇitādhyāya (Mathematics)

This is the most important chapter for the history of mathematics. It contains 72 verses covering:

1. **Arithmetic** (Vyakta-gaṇita): Operations with known quantities — addition, subtraction, multiplication, division, square, square-root, cube, cube-root.
2. **Algebra** (Avyakta-gaṇita): Operations with unknown quantities — equations of one and more unknowns.
3. **Rule of Three** (Trairāśika): Direct and inverse proportion.
4. **Rule of Five, Seven, Nine, Eleven** (Pañcarāśika etc.): Compound proportion.
5. **Mixtures** (Miśraka-vyavahāra): Problems on mixture, alligation, etc.
6. **Series** (Śreḍhī): Arithmetic and geometric progressions.
7. **Geometry** (Kṣetra-vyavahāra): Areas of triangles, quadrilaterals, circles; chords and sines.
8. **Excavations** (Khāta-vyavahāra): Volumes of various solids.
9. **Shadows** (Chāyā): Problems involving shadows and gnomons.

Summary of Brahmagupta’s Mathematical Contributions

Key Mathematical Firsts in the *Brāhmasphuṭasiddhānta*:

1. **Zero as a number:** First systematic rules for arithmetic with zero (Ch. XVIII, verses 30–35).
2. **Negative numbers:** First systematic treatment of positive and negative quantities, including the famous rule “*Riṇayor guṇe dhanam ekam*”—the product of two negatives is positive (Ch. XVIII, v. 33).
3. **Division by zero:** First attempt to define the result of division by zero as *khahara* (infinity)—“*Śūnyena bhaktaś ca khaharaḥ*” (Ch. XVIII, v. 35–36).
4. **Cyclic quadrilateral area:** The formula $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ generalizing Heron’s formula (Ch. XII, vv. 21–26).
5. **Quadratic formula:** First explicit solution procedure for quadratic equations $ax^2 + bx = c$ (Ch. XVIII, vv. 44–48).
6. **Pell’s equation:** The *Bhāvanā* (composition) method for generating solutions to $Nx^2 + 1 = y^2$ (Ch. XVIII, vv. 64–68).
7. **Arithmetic progressions:** Complete theory of A.P.s and G.P.s (Ch. XII, vv. 17–22).
8. **Pulverizer (Kuṭṭaka):** General method for solving linear indeterminate equations $ax + by = c$ (Ch. XVIII, vv. 1–5).

Colophon

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